

ISDA-SIMM Calculation Engine User Guide

A Python program that calculates initial margin (IM) for
uncleared OTC derivatives in USD from a CRIF file

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Quick start

New users only need these pages:
setup, a first run, your own data
and troubleshooting.

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Reference

CRIF rules, the context file, run
options, reading results and
version selection in detail.

Appendix

Error codes · Glossary

Look up an error code from the
screen, or a term you have not
seen before.

Quick start

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SIMM versions 2.5, 2.5A, 2.6, 2.7, 2.7+2412, 2.8+2506, 2.8+2512

Repository <https://github.com/moulinnoirai-cell/ISDA-SIMM-Calculator>

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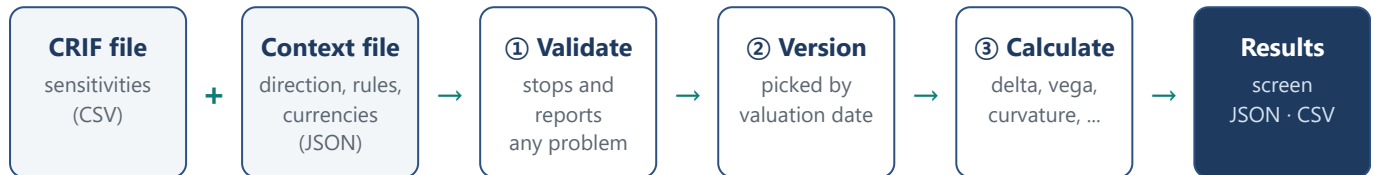
Licence Code and documents: Apache-2.0 / ISDA SIMM parameter values are not covered (rights remain with ISDA)

Quick start 1**At a glance**

Pages 2-5 are enough to run the engine

What does it do?

- Give it a **sensitivity file (CRIF)** for your derivatives and it calculates **initial margin (IM)** with the ISDA SIMM method, in **USD**.
- It checks the input file strictly before calculating. If something is wrong, it does not calculate and tells you **what is wrong and on which line**.
- It selects the **SIMM version** (2.5 to 2.8+2512) that applies to the valuation date.

How it works**What you need**

Item	Description	Where to get it
Python 3.11 or later	Runs the engine. No other packages are needed.	python.org
Engine files	This program.	Download the ZIP from GitHub (Quick start 2)
CRIF file	The sensitivities to calculate (CSV).	Your risk system, or fill in the template
Context file	Run settings (JSON): direction (collect/post), regulation, currency list and so on.	Copy and fill in the template (Quick start 3)

What the results look like

```

SIMM 2.8+2506 | valuation 2026-06-30 | USD | SHADOW ①
Portfolio total: 33,496,812.42 USD ②

[CP_A] total 14,300,634.89 USD ③
  netting set NS_A1 / CSA CSA_1: 14,300,634.89 (add-on 50,000.00) ④
    RatesFX: 7,290,168.27 ⑤
      InterestRate: 1,614,228.80 (Delta 1,559,056.99, Vega 21,989.09, Curvature 33,182.72) ⑥
  
```

① SIMM version, valuation date, currency and the reference-only flag ② total of all counterparties ③ counterparty total ④ netting set (CSA) total and its add-on ⑤ product class ⑥ risk class and its components. All amounts are in USD.

Keep in mind

- **Results are in USD only.** There is no FX conversion: the CRIF `AmountUSD` values are used as they are.
- **No network access.** Your data never leaves the machine during a run.
- Results are **for reference (SHADOW)**. Validate them independently before relying on them. Using ISDA SIMM for margin calculations may require a licence from ISDA (Reference, section 9).

Quick start 2 **First run**

About 10 minutes with the example data · Windows

1 Check that Python is installed

Open **Terminal** (or PowerShell) from the Start menu and type:

```
python --version
```

If it shows **Python 3.11.x** or later, go to the next step.

If you get an error or the Microsoft Store opens

Python is not installed. Download it from python.org and install it. If the installer shows "**Add python.exe to PATH**", tick it. Open a new terminal after installing.

2 Download the engine

1. Go to github.com/moulinnoirai-cell/ISDA-SIMM-Calculator in your browser.
2. Click the green **[Code]** button, then **[Download ZIP]**.
3. Unzip it somewhere (for example `C:\SIMM`). You get a folder named `ISDA-SIMM-Calculator-main`.

3 Open a terminal in the engine folder

Open the `ISDA-SIMM-Calculator-main` folder, right-click an empty area and choose **[Open in Terminal]**. You should be where the `src`, `examples` and `crif` folders are.

4 Prepare the terminal

(type the first line once every time you open a new terminal)

```
$env:PYTHONPATH = "src"
python -m simm versions
```

If you see the list of versions, you are **ready**.

```
2.5      use from COB 2022-12-02  https://www.isda.org/...
...
2.8+2512 use from COB 2026-07-10  https://www.isda.org/...
```

5 Run the example

```
python -m simm run `
  --crif examples\crif_example.csv `
  --valuation-date 2026-06-30 `
  --context examples\context.json
```

If you see this line, it **worked**.

```
SIMM 2.8+2506 | valuation 2026-06-30 | USD | SHADOW
Portfolio total: 33,496,812.42 USD
```

Typing commands

A ``` (backtick) at the end of a line means "continued on the next line". You can paste all lines at once, or type the whole command on one line without the backticks.

The valuation date is 2026-06-30, so 2.8+2506 is used: 2.8+2512 applies from 2026-07-10 (Reference, section 6).

Quick start 3**Calculate your own data**

Prepare two files, run one command

A**Create the CRIF file**

1. Copy `crif\crif_template.csv` into the engine folder and rename it `my_crif.csv`.
2. Open it in Excel. **Keep the first line (column names)** and fill in your data from the second line. The rules are in **Quick start 4**.
3. Use **[Save As] → "CSV UTF-8 (Comma delimited)"**.

Need an example?

Open `examples\crif_example.csv`. It has an example of every kind of row: rates, FX, credit, equity, commodity and add-ons.

B**Create the context file**

Copy `config\context.template.json` into the engine folder, rename it `my_context.json`, open it in Notepad and fill in:

Field	What to enter	Example
<code>direction</code>	<code>collect</code> (margin you receive) or <code>post</code> (margin you give)	<code>"collect"</code>
<code>regulation</code>	Name of the regulation. Must not be empty.	<code>"REG_A"</code>
<code>legal_entity</code>	Your legal entity	<code>"MY_ENTITY"</code>
<code>csa</code>	Name of the CSA (collateral agreement)	<code>"CSA_1"</code>
<code>currency_codes</code>	Every currency in your CRIF. Add only those missing from the default list.	<code>"USD", "EUR", ...</code>
<code>source</code> (in <code>unit_declarations</code>)	For each risk type in your CRIF, the evidence that it was produced in that unit. Risk types you do not use may stay empty.	<code>"Risk system spec v3.2, 4.1"</code>

```
"direction": "collect",
"regulation": "REG_A",
"Risk_IRCurve": { "unit": "PV01_PER_1BP", "source": "Risk system spec v3.2, 4.1" },
```

Note

Do not change the `unit` values. If the `source` of a risk type you use is empty, the run is blocked. This is deliberate: a wrong unit can distort the result badly.

C**Run**

(in a new terminal, run `$env:PYTHONPATH = "src"` first)

```
python -m simm run `
  --crif my_crif.csv `
  --valuation-date 2026-06-30 `
  --context my_context.json `
  --output result.json `
  --summary-csv summary.csv
```

The summary is printed, and `result.json` (full result) and `summary.csv` (a table for Excel) are written to the engine folder. Replace `2026-06-30` with the `ValuationDate` of your CRIF.

Quick start 4

Rules and troubleshooting

The most common mistakes

Six CRIF rules

- 1 Dates as `2026-06-30`, equal to the valuation date you run with.
- 2 No thousands separators: `1000` (yes), `1,000` (no).
- 3 No spaces before or after a cell value.
- 4 Only `AmountUSD` is used. On USD rows `Amount` must equal `AmountUSD`.
- 5 Standard lower-case tenors only: `2w 1m 3m 6m 1y 2y 3y 5y 10y 15y 20y 30y` (credit: `1y 2y 3y 5y 10y`)
- 6 Some cells must be empty depending on the risk type. See the examples below and Reference, section 2.

Common risk type examples

ProductClass	RiskType	Qualifier	Bucket	Label1	Label2	Amount	AmountCurrency	AmountUSD
RatesFX	Risk_IRCurve	USD	1	5y	OIS	25000	USD	25000
RatesFX	Risk_FX	EUR				900000	USD	900000
RatesFX	Risk_FXVol	EURUSD		1y		30000	USD	30000
Equity	Risk_Equity	ISIN:KR0000000036	1			200000	USD	200000
Credit	Risk_CreditQ *	ISIN:XS0000000016	3	5y		15000	USD	15000

Leave blank cells empty. The first five columns (ValuationDate to IMModel) are omitted here. * Credit rows also need `PaymentCurrency`.

When you get stuck

Message on the screen	What to do
No module named simm	Check that the terminal is open in the engine folder and run <code>\$env:PYTHONPATH = "src"</code> again.
DATE_MISMATCH	Make the CRIF date and <code>--valuation-date</code> the same.
UNIT_EVIDENCE_REQUIRED	Fill in the <code>source</code> of that risk type in the context file.
SCOPE_REQUIRED	Fill in <code>legal_entity</code> and <code>csa</code> in the context file.
DIRECTION_REQUIRED	Set <code>direction</code> to "collect" or "post".
INVALID_AMOUNT	Remove commas or fill empty amount cells.
WHITESPACE	Remove spaces around the cell values on that line.
INVALID_TENOR	Use a standard tenor in Label1 (rule 5).
INVALID_CURRENCY	Add the currency to <code>currency_codes</code> in the context file.
Inconsistent CSV field count	Lines have different numbers of cells. Put values that contain commas in double quotes.

Reading an error message

`CRITICAL DATE_MISMATCH row:5` means "the date on line 5 of the CSV file (Excel row 5) is different". Only the first 20 issues are shown; add `--output issues.json` to the command to get all of them. Other error codes are in

Appendix A.

Reference

Everything the quick start does not cover, in detail

1. Installation and environment

1.1 Requirements

Item	Details
Python	3.11 or later. With several versions installed you can pick one, for example <code>py -3.11</code> .
Extra packages	None. Only reading an Excel (XLSX) CRIF directly needs <code>pip install openpyxl</code> .
Operating system	Windows, macOS and Linux.

1.2 Folder layout

Folder / file	Contents
<code>crif\</code>	CRIF template (<code>crif_template.csv</code>) and the format description (<code>CRIF_FORMAT.md</code>)
<code>config\</code>	Context file template (<code>context.template.json</code>)
<code>examples\</code>	Synthetic example CRIF and context file, ready to run (not real trades)
<code>parameters\</code>	Risk weights, correlations and other values per SIMM version (transcribed from ISDA's public documents)
<code>src\</code> , <code>tests\</code>	Program code and tests
<code>docs\</code>	This guide (English and Korean) and the methodology notes

1.3 Downloading with git (optional)

With git you can clone instead of downloading the ZIP. The folder is then named `ISDA-SIMM-Calculator`.

```
git clone https://github.com/moulinnoirai-cell/ISDA-SIMM-Calculator.git
```

1.4 Preparing each kind of terminal

The engine code is in the `src` folder. Tell Python where it is every time you open a terminal.

Terminal	Preparation command	Line continuation character
Windows PowerShell / Terminal	<code>\$env:PYTHONPATH = "src"</code>	<code>`</code> (backtick) at the end of the line
Windows Command Prompt (cmd)	<code>set PYTHONPATH=src</code>	<code>^</code> at the end of the line
macOS / Linux	<code>export PYTHONPATH=src</code>	<code>\</code> at the end of the line

On macOS and Linux use `/` in paths (for example `examples/crif_example.csv`). Some systems use `python3` instead of `python`.

1.5 Installing (optional)

Running `pip install -e .` in the engine folder gives you a `simm` command that works without the preparation step.

- **Keep the `-e`**. The engine reads the `parameters` folder of the downloaded copy, so a regular install (`pip install .`) does not work.
- Installing downloads build tools from the internet. If that fails (for example behind a corporate network), use the no-install method in 1.4.

1.6 Checking that everything works

Run the built-in tests. **Ran 60 tests ... OK** at the end means all is well.

```
python -m unittest discover -s tests -t .
```

2. Writing the CRIF file

CRIF (Common Risk Interchange Format) is the standard format for exchanging SIMM sensitivities. Each line is the sensitivity to one risk factor. The engine reads CSV, TSV and single-sheet XLSX files.

2.1 Columns

Column	Required	Description	Example
ValuationDate	Recommended	Valuation date (YYYY-MM-DD). Must equal the date given on the command line.	2026-06-30
Counterparty	Yes *	Counterparty identifier	CP_A
NettingSet	Yes *	Netting set (CSA portfolio). Netting sets are added up, never offset .	NS_A1
TradeID	Optional	Trade identifier. Keeps identical rows of different trades apart.	T001
IMModel	Optional	Only SIMM is allowed. Schedule or anything else blocks the run.	SIMM
ProductClass	Yes	RatesFX , Credit , Equity or Commodity	RatesFX
RiskType	Yes	See the table in 2.2	Risk_IRCurve
Qualifier	Yes	Currency, issuer, index and so on	USD
Bucket	Yes (may be blank)	Bucket of the risk type	1
Label1	Yes (may be blank)	Tenor or option expiry	5y
Label2	Yes (may be blank)	Curve type and similar	OIS
Amount	Yes	Sensitivity in AmountCurrency	25000
AmountCurrency	Yes	Currency of Amount (ISO code)	USD
AmountUSD	Yes	Sensitivity in USD. This is the only amount used.	25000
PaymentCurrency	Credit only	Payment currency of credit delta and vega rows. The same issuer in different payment currencies gives different risk factors.	USD
GroupName	Credit non-qualifying only	Group name of non-qualifying credit. The correlation depends on whether the group is the same.	CMBX

* Counterparty and NettingSet may be left out and taken from the context file defaults. Your legal entity (LegalEntity) and CSA work the same way. The template has no LegalEntity or CSA column, so put defaults in the context file.

Optional columns not in the template

Column	Description
LegalEntity , CSA	Use them when the legal entity or CSA differs between rows.
SIMMVersion	If present, it must equal the selected version.
CollectRegulations , PostRegulations	If present, rows whose list does not include the context regulation are excluded. The number of excluded rows is recorded in the result.

Column	Description
Unit , SensitivityUnit	If present, they must equal the unit declared in the context file.

2.2 Rules for each risk type

Tenors (Label1) are standard lower-case values only: 2w 1m 3m 6m 1y 2y 3y 5y 10y 15y 20y 30y . Credit tenors are 1y 2y 3y 5y 10y . Other tenors (for example 7y) are rejected rather than re-gridded. Cells marked "blank" must be empty.

RiskType	Qualifier	Bucket	Label1	Label2	Amount unit
Risk_IRCurve	Currency	1 regular, 2 low vol (JPY), 3 high vol	Tenor	OIS , Libor1m , Libor3m , Libor6m , Libor12m (USD also Prime , Municipal)	PV01 per 1bp
Risk_Inflation	Currency	blank	blank	blank	PV01 per 1bp
Risk_XCcyBasis	Currency	blank	blank	blank	PV01 per 1bp
Risk_IRVol	Currency	blank	Option expiry	blank	Vol-weighted vega ($\sigma \times \partial V / \partial \sigma$)
Risk_InflationVol	Currency	blank	Option expiry	blank	Vol-weighted vega
Risk_CreditQ	ISIN: + 12-character ISIN	1 - 12 , Residual	Credit tenor	blank or Sec	CS01 per 1bp
Risk_CreditVol	Issuer	1 - 12 , Residual	Credit tenor	blank	Vol-weighted vega
Risk_BaseCorr	Index family (e.g. CDX IG)	blank	blank	blank	BC01 per 1 correlation point
Risk_CreditNonQ	Tranche	1 , 2 , Residual	Credit tenor	blank	CS01 per 1bp
Risk_CreditVolNonQ	Tranche	1 , 2 , Residual	Credit tenor	blank	Vol-weighted vega
Risk_Equity	Buckets 1-10, Residual: ISIN: + ISIN Buckets 11-12: any name	1 - 12 , Residual	blank	blank	Delta per 1% relative move
Risk_EquityVol	As Risk_Equity	1 - 12 , Residual	Option expiry	blank	Vega per 1 vol point (σ applied by the engine)
Risk_Commodity	Commodity	1 - 17	blank	blank	Delta per 1% relative move
Risk_CommodityVol	Commodity	1 - 17	Option expiry	blank	Vega per 1 vol point
Risk_FX	Currency (USD rows are allowed and excluded)	blank	blank	blank	Delta per 1% relative move
Risk_FXVol	Currency pair, 6 letters (EURUSD = USDEUR)	blank	Option expiry	blank	Vega per 1 vol point

2.3 Add-on and multiplier rows

Additional margin from section L of the SIMM methodology. It applies to the row's netting set. Leave `Bucket`, `Label1` and `Label2` blank.

RiskType	ProductClass	Qualifier	AmountUSD
<code>Param_ProductClassMultiplier</code>	Same as Qualifier	Product class	Multiplier of 1 or more (e.g. 1.2)
<code>Param_AddOnNotionalFactor</code>	blank	Product name	Add-on factor in percent (5 = 5%)
<code>Notional</code>	blank	Product name	Trade notional (absolute values are summed)

- Add-on = sum of absolute notionals × factor ÷ 100. `Param_AddOnNotionalFactor` and `Notional` rows must pair up by product name.
- On these rows `Amount` must equal `AmountUSD`.
- A fixed add-on amount is not a CRIF row; it is given through Python configuration (section 7).

```
2026-06-30,CP_A,NS_A1,,SIMM,RatesFX,Param_ProductClassMultiplier,RatesFX,,,1.2,USD,1.2,,
2026-06-30,CP_A,NS_A1,,SIMM,,Param_AddOnNotionalFactor,ProductX,,,5,USD,5,,
2026-06-30,CP_A,NS_A1,T018,SIMM,,Notional,ProductX,,,1000000,USD,1000000,,
```

2.4 Working in Excel

- **File type:** save as "CSV UTF-8 (Comma delimited)". Other CSV types may garble non-ASCII text or fail to load.
- **Dates:** open the saved file in Notepad and check that Excel did not turn `2026-06-30` into something like `6/30/2026`. If dates were saved as numbers such as `46203`, add `--excel-serial-dates` when running.
- **Numbers:** `1000` and `1E+5` are fine; thousands separators are not.
- **Loading XLSX directly:** one sheet only and no formulas (paste as values). Needs `openpyxl`.

2.5 Rows in currencies other than USD

- `AmountCurrency` may be EUR and so on. `AmountUSD` must then hold the **USD value converted by whoever produced the CRIF**. The engine does not check FX rates; it only records a `FX_RATE_UNVERIFIED` warning.
- If `Amount` and `AmountUSD` have different signs, or only one of them is zero, the row is rejected.

3. Writing the context file

The template is `config/context.template.json`; a filled-in example is `examples/context.json`.

Field	Description	Example
<code>direction</code>	<code>collect</code> (margin you receive) or <code>post</code> (margin you give). It does not change the amounts; it labels the result and selects the regulation column.	"collect"
<code>regulation</code>	Name of the regulation; must not be empty. If the CRIF has <code>CollectRegulations</code> / <code>PostRegulations</code> columns, rows are filtered by this name.	"REG_A"
<code>crif_profile</code>	Leave as it is (<code>public_rds_1_36_methodology_2_8</code>).	
<code>currency_codes</code>	Accepted currencies. Must include every currency in the CRIF (<code>AmountCurrency</code> , rates and FX <code>Qualifier</code> , <code>PaymentCurrency</code>).	["USD","EUR","KRW"]
<code>counterparty</code>	(Optional) default when the CRIF has no <code>Counterparty</code> column. Not in the template; add it if needed.	"CP_A"
<code>legal_entity</code>	Default when the CRIF has no <code>LegalEntity</code> column	"MY_ENTITY"

Field	Description	Example
<code>csa</code>	Default when the CRIF has no <code>CSA</code> column	"CSA_1"
<code>netting_set</code>	Default when a CRIF <code>NettingSet</code> value is blank	"DEFAULT"
<code>unit_declarations</code>	For each risk type, the unit your CRIF uses (<code>unit</code>) and the evidence for it (<code>source</code>)	see below

Filling in unit_declarations

A wrong sensitivity unit (for example per 1% instead of per 1bp) can make the result 100 times too large or too small. So for each risk type you record the evidence that your CRIF was produced in that unit.

```
"Risk_IRCurve": { "unit": "PV01_PER_1BP", "source": "Risk system CRIF specification v3.2, section 4.1" }
```

- `unit`: keep the template value. Changing it causes a `UNIT_MISMATCH` error.
- `source`: the document or check that confirms the unit. If it is empty for a risk type in your CRIF, the run is blocked. Risk types you do not use may stay empty.

Credential guard

The run is refused if the context file has a field name containing `password`, `token`, `secret`, `api_key`, `cookie` or `credential`.

4. Run command and options

```
python -m simm run --crif <CRIF file> --valuation-date <YYYY-MM-DD> --context <context file> [options]
```

Option	Description
<code>--crif</code> file	(Required) CRIF file path
<code>--valuation-date</code> date	(Required) Valuation date. Must equal the CRIF <code>ValuationDate</code> .
<code>--context</code> file	(Required) Context file path
<code>--output</code> result.json	Writes the result as JSON. If the run is blocked, it writes the full list of issues instead.
<code>--include-trace</code>	Adds the full calculation trace to <code>--output</code> . The file gets large.
<code>--summary-csv</code> summary.csv	Writes the counterparty / netting set / product class / risk class table as CSV.
<code>--version</code> 2.7	Sets the SIMM version explicitly. A warning is shown if it differs from the scheduled one.
<code>--netting-set-column</code> name	Uses another CRIF column (e.g. <code>PortfolioId</code>) as the netting set. Not allowed if the CRIF already has a <code>NettingSet</code> column.
<code>--excel-serial-dates</code>	Converts Excel date numbers (e.g. 46203) to <code>2026-06-30</code> format.
<code>--quiet</code>	Does not print the summary. Use it when you only need the files.

Options that change the input, such as `--netting-set-column` and `--excel-serial-dates`, are recorded in `input_transformations` in the JSON result.

Exit codes

Code	Meaning
0	Calculation succeeded
1	Run error (file not found, wrong format, unsupported date, ...)

Code	Meaning
2	Blocked by validation; the issues are listed on screen.

Running several dates at once (PowerShell)

```
foreach ($d in "2026-03-31", "2026-06-30") {
    python -m simm run --crif "data\crif_$d.csv" --valuation-date $d `
    --context my_context.json --output "out\result_$d.json" --quiet
    if ($LASTEXITCODE -ne 0) { Write-Host "$d failed (code $LASTEXITCODE)" }
}
```

Create the `data` and `out` folders first.

5. Reading the results

5.1 How amounts add up

Portfolio total	Simple sum of the counterparty totals
└ Counterparty	Simple sum of its netting set totals
└ Netting set	Simple sum of its product class IMs plus add-ons. Netting sets never offset each other.
└ Product class	Risk class IMs combined with correlations (square root of the sum of squares) × multiplier
└ Risk class	Delta + Vega + Curvature (+ BaseCorr for qualifying credit)

5.2 Screen output (full example)

```
SIMM 2.8+2506 | valuation 2026-06-30 | USD | SHADOW
Portfolio total: 33,496,812.42 USD

[CP_A] total 14,300,634.89 USD
netting set NS_A1 / CSA CSA_1: 14,300,634.89 (add-on 50,000.00)
RatesFX: 7,290,168.27
InterestRate: 1,614,228.80 (Delta 1,559,056.99, Vega 21,989.09, Curvature 33,182.72)
FX: 5,697,557.53 (Delta 5,545,277.27, Vega 94,228.44, Curvature 58,051.83)
Credit: 6,960,466.62
InterestRate: 91,500.00 (Delta 91,500.00)
CreditQualifying: 1,268,337.75 (Delta 1,228,592.85, Vega 8,400.00, Curvature 2,544.89, BaseCorr 28,800.00)
CreditNonQualifying: 6,111,433.76 (Delta 6,108,150.29, Vega 2,520.00, Curvature 763.47)

[CP_B] total 19,196,177.54 USD
netting set NS_B1 / CSA CSA_1: 18,708,177.54 (add-on 0.00)
Equity: 12,135,837.73
InterestRate: 198,000.00 (Delta 198,000.00)
Equity: 11,559,433.25 (Delta 10,440,818.93, Vega 470,673.89, Curvature 647,940.42)
FX: 1,775,000.00 (Delta 1,775,000.00)
Commodity: 6,572,339.80
Commodity: 6,572,339.80 (Delta 6,046,261.65, Vega 379,453.70, Curvature 146,624.45)
netting set NS_B2 / CSA CSA_1: 488,000.00 (add-on 0.00)
RatesFX: 488,000.00
InterestRate: 488,000.00 (Delta 488,000.00)
```

A product class IM (e.g. RatesFX 7,290,168.27) is smaller than the simple sum of its risk class IMs (1,614,228.80 + 5,697,557.53) because of diversification between risk classes. It also includes the multiplier of 1.2.

5.3 CSV output (`--summary-csv`)

One line per counterparty × netting set × product class × risk class. The file is UTF-8 with a BOM, so Excel opens it correctly.

Columns	Contents
<code>counterparty</code> , <code>netting_set</code> , <code>csa</code> , <code>direction</code>	Scope
<code>product_class</code> , <code>risk_class</code>	Classification
<code>delta</code> , <code>vega</code> , <code>curvature</code> , <code>basecorr</code>	IM per component
<code>risk_class_margin</code>	Stand-alone risk class IM
<code>product_margin</code> , <code>netting_set_total</code> , <code>counterparty_total</code>	Higher-level totals, repeated on several lines. Do not add them up again in Excel.

If the numbers look very long

The engine calculates with 50-digit precision and writes unrounded values. Format the cells in Excel as "Number, 2 decimal places".

5.4 JSON output (`--output`)

Location	Contents
<code>metadata</code>	Input file and its hash (<code>crif_sha256</code>), valuation date, selected version and warning, parameter hash, input transformations, validation status, warning counts, and rows read, used and excluded
<code>result.counterparties[]</code>	Results in the order counterparty → <code>netting_sets[]</code> → <code>products[]</code> → <code>risk_classes[]</code> , with amounts at every level
<code>risk_classes[].contribution</code>	The product class IM split by risk class. Within a product class they add up to the product class IM.
<code>result.trace</code>	Only with <code>--include-trace</code> . Every intermediate value from the CRIF rows to the totals, with the SIMM methodology paragraph, so you can follow how any number was produced.

6. SIMM version selection

ISDA announces each version with a "use from close of business (COB)" date. The engine uses the latest version whose date is on or before the valuation date.

Version	Use from (COB)	Valuation dates it is used for
2.5	2022-12-02	2022-12-02 to 2023-07-13
2.5A	2023-07-14	2023-07-14 to 2023-11-30
2.6	2023-12-01	2023-12-01 to 2024-12-05
2.7	2024-12-06	2024-12-06 to 2025-07-10
2.7+2412	2025-07-11	2025-07-11 to 2025-12-04
2.8+2506	2025-12-05	2025-12-05 to 2026-07-09
2.8+2512	2026-07-10	2026-07-10 onwards

To see which version applies to a date:

```
python -m simm versions --date 2025-03-31
```

```
{ "version": "2.7", "use_from_cob": "2024-12-06", "valuation_date": "2025-03-31",
  "beyond_verified_schedule": false, "schedule_verified_through": "2026-10-06", "warning": null }
```

- **Before 2022-12-02:** not supported (No SIMM package scheduled error).
- **After 2026-10-06:** calculated with the latest version and flagged with a warning, because ISDA may have published a newer version since. Check the ISDA website; if there is a new version, add it as described in section 8.
- **Setting the version yourself:** if you agreed a different version with your counterparty, use `--version`. A warning is recorded on screen and in the JSON.

```
SIMM 2.7 | valuation 2026-06-30 | USD | SHADOW
WARNING: Version override 2.7 differs from scheduled 2.8+2506
```

7. Using the engine from Python

You can call the engine from Python code instead of the command line. A **fixed add-on amount** can only be given this way.

```
import json
from simm.crif import read_crif, validate
from simm.engine import Engine
from simm.parameters import ParameterPackage
from simm.versioning import select_version

valuation_date = '2026-06-30'
context = json.load(open('examples/context.json', encoding='utf-8'))
context['valuation_date'] = valuation_date

selection = select_version(valuation_date)          # select the version
package = ParameterPackage(selection['version'])    # load parameters (hash checked)
headers, rows = read_crif('examples/crif_example.csv')
validation = validate(headers, rows, package, context)

if validation.blocked:
    for issue in validation.issues:
        print(issue.severity, issue.code, issue.row_id, issue.message)
else:
    config = {'fixed_addons': [{
        'scope': {'counterparty': 'CP_B', 'legal_entity': 'ENTITY_1', 'netting_set': 'NS_B2',
                  'csa': 'CSA_1', 'direction': 'collect', 'regulation': 'EXAMPLE_REGULATION'},
        'amount_usd': '10000',
        'source': 'CSA side letter 2026-01-15'}}]
    result = Engine(package).calculate(validation, config)
    print(result['simm_version'], result['total_simm'])
```

- **scope**: all six fields must match a **scope** in the JSON result. If no netting set matches, an error is raised.
- **amount_usd**: a USD amount of 0 or more. Pass it as a string to keep full precision. **source** (the agreement reference) must not be empty.
- Running this example raises NS_B2 from 488,000 to 498,000 and the total to 33,506,812.42.

8. Adding a new SIMM version

When ISDA publishes a new version and the formulas (methodology sections B-C) are unchanged, adding its parameters is enough.

1 Create the folder

Create `parameters\simm_<version>`, replacing `.` and `+` with `_` (e.g. 2.8+2512 → `simm_2_8_2512`).

2 Write parameters.json

Copy the file of an existing version and change the values according to the new ISDA methodology document. Update `version` and `sources` as well.

3 Compute the hash

Put the file's SHA-256 into `sha256` of `manifest.json`, in lower case.

```
(Get-FileHash parameters\simm_X_Y\parameters.json).Hash.ToLower() # Windows PowerShell
shasum -a 256 parameters/simm_X_Y/parameters.json                 # macOS / Linux
```

4 Add it to the schedule

Append `version`, `use_from_cob`, `announcement` and `methodology_pdf` to `versions` in `parameters\version_schedule.json`, and set `verified_through` to the date you checked ISDA's announcements.

5 Check

Run the tests and check that `python -m simm versions` lists the new version.

When parameters are not enough

If the formulas change or a new `methodology_id` appears, the engine code must change too (for example `SUPPORTED_METHODOLOGIES` in `src\simm\parameters.py`). Parameter files are checked when loaded: hash, symmetry and range of correlation matrices, and concentration thresholds.

9. Notices and limitations

- **ISDA rights:** ISDA SIMM® is a registered trademark and intellectual property of ISDA and is subject to U.S. Patent No. 10,515,410. The values in `parameters` are transcribed from ISDA's public documents and are not covered by this project's Apache-2.0 licence.
- **ISDA licence:** using SIMM for margin calculations may require a licence from ISDA. See the ISDA SIMM licensing FAQ ([isda.org](https://www.isda.org)).
- **Validation:** the parameters were checked against independent transcriptions and the formulas against an independent re-implementation on synthetic portfolios. ISDA's official unit tests (licensed material) were not run. Details are in `docs\METHODOLOGY_NOTES.md`.
- **CRIF standard:** based on the public Risk Data Standards 1.36. The `PaymentCurrency` and `GroupName` columns and the fixed add-on configuration are extensions of this engine; check that they match your CRIF producer.
- **Not covered:** FX conversion, thresholds, minimum transfer amounts, collateral valuation and regulatory scope decisions.
- **Licence:** code, documents and templates are Apache-2.0. You may use, modify, redistribute and use them commercially without permission, provided you keep LICENSE and NOTICE and mark the files you change.

10. FAQ

Q. I need results in another currency, such as KRW.

The engine reports USD only. Multiply the result by the FX rate you need. Which rate (date, source) to use should follow your agreement with the counterparty.

Q. Can two CSAs with the same counterparty be calculated together?

Each CSA (netting set) is calculated separately and the results are added. To offset them, give rows of both CSAs the same `NettingSet` value, but only if the legal agreements actually allow it.

Q. My CRIF contains trades under Schedule IM.

Rows whose `IMModel` is not `SIMM` block the run. Move Schedule rows to a separate file.

Q. I only get warnings (WARNING). Is that OK?

Yes, the calculation runs. `LATEST_RDS_UNVERIFIED` always appears. `NETTED_ROWS` means several rows of the same risk factor were added together; check that this is intended.

Q. My result differs slightly from my counterparty's or another system's.

Check, in this order: ① the same SIMM version ② the same CRIF (`crif_sha256`) ③ the same netting sets ④ the same FX rates for non-USD rows ⑤ the same rounding. Comparing intermediate values with `--include-trace` helps find where the difference starts.

Appendix A

All error codes

CRITICAL and REVIEW_REQUIRED block the run; WARNING does not

Files and settings

Code	Cause	Fix
MISSING_COLUMNS	A required column is missing	Use the template column names exactly (case-sensitive).
DIRECTION_REQUIRED	<code>direction</code> is not collect/post	Replace the template's "collect post" with one of them.
REGULATION_REQUIRED	<code>regulation</code> is empty	Enter a regulation name.
PROFILE_REQUIRED	Wrong <code>crif_profile</code>	Keep the template value.
CURRENCY_REGISTRY_REQUIRED	<code>currency_codes</code> is empty	Enter the currency list.
UNIT_EVIDENCE_REQUIRED	<code>unit</code> / <code>source</code> empty for a risk type in use	Enter the evidence (section 3).
UNIT_MISMATCH	Declared unit differs from the convention	Restore the template <code>unit</code> and check the CRIF units.
SCOPE_REQUIRED	Counterparty, legal entity, netting set or CSA is empty	Fill the column or set a context default.
EMPTY_PORTFOLIO	The CRIF has no data rows	Check the file contents.

Row contents

Code	Cause	Fix
DATE_MISMATCH	CRIF date $\neq$ <code>--valuation-date</code>	Make the dates equal.
INVALID_DATE	Date not in <code>YYYY-MM-DD</code> format	Fix the format or use <code>--excel-serial-dates</code> .
WHITESPACE	Spaces around a cell value	Remove them at the source.
EXACT_DUPLICATE	The same row appears twice	Risk of double counting. If they are different trades, set <code>TradeID</code> .
UNSUPPORTED_MODEL	<code>IMModel</code> is not SIMM	Move Schedule rows to another file.
VERSION_MISMATCH	<code>SIMMVersion</code> column $\neq$ selected version	Check the column or <code>--version</code> .
UNKNOWN_RISK_TYPE	Unsupported RiskType	Check the spelling.
INVALID_PRODUCT_CLASS	Wrong product class	Use RatesFX, Credit, Equity or Commodity.
MISSING_QUALIFIER	Qualifier is empty	Enter a value.
INVALID_AMOUNT	Not a number (empty, commas, ...)	Use the <code>1000</code> format.
INVALID_CURRENCY	<code>AmountCurrency</code> not in the currency list	Add it to <code>currency_codes</code> .
USD_AMOUNT_MISMATCH	USD row with Amount $\neq$ AmountUSD	Make them equal.
FX_SIGN_OR_ZERO	Non-USD row: different signs, or only one is zero	Check the converted value.
INVALID_QUALIFIER_CURRENCY	Rates/FX qualifier not in the currency list	Check the currency code.
INVALID_FX_PAIR	Wrong FX vol currency pair	Use the <code>EURUSD</code> format.
IR_BUCKET_MISMATCH	Rates bucket does not fit the currency	Use the value in the message (1/2/3).
CURVE_MAPPING	Rates Label2 not in the allowed list	Map it to OIS, Libor3m and so on.

Code	Cause	Fix
INVALID_BUCKET	Credit/equity/commodity bucket out of range	Check the ranges in 2.2.
UNUSED_BUCKET / UNUSED_LABEL1	Value in a cell that must be blank	Empty the cell.
INVALID_TENOR	Tenor not allowed	Re-produce on standard tenors.
UNCONFIRMED_LABEL2	Label2 on a non-rates risk type	Empty it (except Sec on CreditQ).
PAYMENT_CURRENCY_REQUIRED	Credit row without payment currency	Fill PaymentCurrency .
NQ_GROUP_REQUIRED	Non-qualifying credit row without group	Fill GroupName .
QUALIFIER_MAPPING	CreditQ or equity (buckets 1-10, Residual) qualifier is not an ISIN	Use the ISIN:US0378331005 format.
INCONSISTENT_BUCKET	Same qualifier with different buckets	Use one bucket.

Add-on rows

Code	Cause	Fix
ADDON_LABELS	Bucket/Label filled on an add-on row	Empty them.
INVALID_MULTIPLIER	Multiplier below 1 or wrong product class	Use 1 or more and a valid product class.
MULTIPLIER_CLASS_MISMATCH	ProductClass ≠ Qualifier	Make them equal.
NEGATIVE_ADDON	Negative add-on factor	Use 0 or more.
PARAM_AMOUNT_MISMATCH	Add-on row with Amount ≠ AmountUSD	Make them equal.
FIXED_ADDON_ENCODING	Fixed add-on given as a CRIF row	Use the Python fixed_addons setting (section 7).

Warnings (the run continues)

Code	Meaning
LATEST_RDS_UNVERIFIED	Always shown. Not checked against the latest licensed Risk Data Standards and ISDA's official tests.
FX_RATE_UNVERIFIED	The AmountUSD of a non-USD row was used without checking.
NETTED_ROWS	Several rows of the same risk factor were added together.

Run errors (exit code 1)

Message	Cause and fix
Inconsistent CSV field count	Lines have different numbers of cells. Quote values that contain commas.
Duplicate column names	A column name appears twice.
Column whitespace ...	A column name has spaces around it.
Select one CRIF worksheet ...	The XLSX has several sheets. Keep only one.
Excel formula cells are not accepted ...	The XLSX contains formulas. Paste as values.
Supported inputs: ...	Unsupported file type (e.g. .xls). Convert to CSV or XLSX.
No SIMM package scheduled for ...	Date before 2022-12-02.
Parameter package hash mismatch	A parameters file was changed. Download it again.
Credentials are prohibited ...	The context file has a credential-like field name. Remove it.

Message	Cause and fix
CRIF already has NettingSet ...	--netting-set-column used although a NettingSet column exists. Use only one.
Notional/factor product coverage differs ...	Add-on factor rows and Notional rows do not pair up by product name.
Conflicting product class multipliers	Different multipliers for the same netting set and product class.

Appendix B Glossary

Term	Meaning
SIMM	ISDA's Standard Initial Margin Model for uncleared OTC derivatives
IM (initial margin)	Collateral exchanged up front to cover losses if a counterparty defaults
CRIF	Common Risk Interchange Format, the standard file format for SIMM sensitivities
Sensitivity	How much a trade's value changes when a market variable such as a rate moves slightly
Delta / Vega / Curvature	First-order price sensitivity / volatility sensitivity / non-linear effect of large moves
BaseCorr	Sensitivity of credit index tranches to base correlation
bp, PV01, CS01	bp = 0.01%. PV01 and CS01 = value change for a 1bp move in rates or credit spreads
Netting set	A group of trades that can be offset against each other, usually one CSA
CSA	Credit Support Annex, the collateral agreement
COB	Close of business, the valuation point in time
Product class	The four product groups RatesFX, Credit, Equity and Commodity
Risk class	The six risk groups InterestRate, CreditQualifying, CreditNonQualifying, Equity, Commodity and FX
Qualifier / Bucket / Label	Cells that identify a risk factor: currency or issuer / group within the risk class / tenor or curve
Tenor	Maturity point (e.g. 5y = 5 years)
Add-on / Multiplier	Additional margin added to / factor applied to the calculated SIMM
ISIN	12-character international securities identification number
SHADOW	Marks a reference calculation, not an official margin figure